Lloyds Banking Group: Analysis of retail-banking customer data

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***Abstract*— The project deals with two synthetic datasets provided by Lloyds Banking Group (LBG) containing bank transactional data that is very similar to real-world data. Performing an exploratory data analysis on these datasets can help answer several compelling questions that could reveal useful insights for decision-makers, such as: understanding customer behaviour, average customer spending, identifying peak spending months, which shops have the highest spending and whether customers that spend more tend to transfer money to other people, who also spend more. Our main contribution is analysing two synthetic datasets, consisting of 13234863 million transactional records and 174601 transactional records, respectively. This dataset was combined with insights from the analysis, which included a dashboard, representing a detailed customer transaction for money payed into the customer's account and money paid out of the account, along with details of the counter-party to the transaction. We also applied unsupervised and supervised learning techniques, including data clustering and building a decision tree classification model, to classify customers into four different groups with similar characteristics, based on the customer's spending density and transfer frequency. We found that the customers with the highest spending frequency in terms of transactions had been spending their money in bars or pubs. Moreover, we found that customer spending increased at the start of the year (January) and at the end of the year (December). We concluded that banks should classify customers according to their profits and loyalty to the bank when recommending suitable financial products to them with corresponding service quality. To create this recommendation engine, we have built a classification model that divides all customers into four groups: the ordinary customer, loyal customer, potential valuable customer and priority customer.**

Keywords — Exploratory Data Analysis, Data Mining, Classification, Clustering, Banking sector, central banks, finance.

I. Introduction

A vast amount of transactional bank data is generated daily and, essentially, refers to everything related to customer activity; their behaviour and personality is shown in their transactions. Therefore, conducting a proper analysis on this data is required in order to gain useful insights to help make better-informed decisions.

Most financial products and services are standard, making it difficult for banks to meet the unique needs of their customers. By understanding client behavioural trends and preferences, banks would be able to formulate an affective marketing strategy. The important question is, how could Lloyds Banking Group (LBG) take advantage of these huge amounts of data to help them increase their profits, as well as customer loyalty? Detecting fraud is another critical aspect to consider in the banking sector and this can be done by clustering customers and monitoring outliers and suspicious behaviours.

Exploratory Data Analysis (EDA) aims at analysing data sets encompass a broader overview of their main characteristics. This is done by using statistical graphics and other data visualisation methods. Although the available datasets do not have a customer class label, unsupervised learning addresses this problem by using unlabelled data and adding a class label for each customer, to learn classification models.

In this research paper, we employ two well-known supervised and unsupervised learning approaches, k-means clustering and decision tree classification, to cluster customers into different groups, sharing similar characteristics, before classifying them.

The rest of the paper is organised as follows: related work is presented in Section 2. In Section 3, we describe our dataset and the preprocessing techniques used to render the data ready for analysis. The methods used to derive our results are described in Section 4. Our results are described in Section 5, future work and conclusions in Section 6.

II. Related Work

Our related is derived from different areas: central banking studies, fraud detection, classification models, customer retention in the banking sector and data mining in the banking sector. Wei Wei, Jinjiu Li, Longbing Cao, Yuming Ou and Jiahang Chen [1] propose a novel algorithm called ‘ContrastMiner' that effectively detects and distinguishes fraudulent activity from genuine customer behaviour. It does this by building a contrast vector for each customer transaction, based on the customer’s historical behavioural sequence and outlines the differentiating rate of each current transaction against the customer’s behavioural preference. The results of the ‘ContrastMiner' algorithm on real online banking data show high fraud detection accuracy and it works better than existing fraud detection methods and systems in both efficiency and accuracy. In addition, the algorithm can be merged with the existing banking fraud detection system.

Bholat [2], explains why 'Big Data' is likely to become of paramount importance to central banks in changing both internal operations and external economic and financial systems. 'Big Data’ tools and techniques have had less of an impact on financial services than they have had on other sectors of the economy. Nevertheless, the situation appears to be changing rapidly. Having standardised granular data is

critical for central banks as they link to legal considerations to expand the scope of granular data collections, as well as the advancement of inductive analytical approaches to enhance the conventional deductive approaches used by central banks. Bholat concludes that ‘Big Data' will play a major role in changing how central banks operate and how careful analysis of their data could enhance productivity and profits. Hence, there is the need for central banks to follow an effective strategic approach to data analysis to better understand their customers. This would help them to achieve their customers’ needs by thoroughly investigating their daily transactional data.

Chitra and Subashini apply a data mining solution to investigate the customer churn problem which is considered one of the major problems in the banking sector [3]. They have built a decision tree classification technique that generates a tree and set of rules representing the model of different classes from a given dataset.

Similarly to us, they have used a method to build a decision tree called ‘CART'. ‘CART' builds a binary decision tree by splitting the records at each node, according to a function of a single attribute. The model then predicts the future status of the customers as either ‘active’ or ‘churn'. Comparing their study approach to the dataset that we have for Lloyds Banking Group customers, we applied their customer classification technique to classify customers into four groups: ordinary customers, loyal customers, potential valuable customers and priority customers. Our classification was based on customer spending density and the number of transactions. In their study, they have four sets of data variables to consider: customer behaviour, customer perceptions, customer demographics and macro environment variables. Their approach discovers the churn patterns within past churners’ behaviour and uses this knowledge to assign a potential churn score rating to current customers. The study could be useful to bank managers to help form a strategy for customer retention activities, in collaboration with the ‘Customer Relationship Management’ department.

Subia and Galapon [4] present a step-by-step approach in the creation of the sample model for predicting default risk.

They collect and save bank data in an ARFF format, which is composed of essential attributes, such as credit history, existing credits, purpose, amount of money, gender, age, job and class. Credit history includes the following factors: 'all paid’, 'critical existing credit’,'delayed previously’, 'existing paid/not yet due’. Job includes factors such as: 'high qualif/self emp/mgmt’, ‘skilled’,’unskilled’,’retired/unemployed’. The class refers to users’ current credit status, whether it is paid accordingly or inconsistently. They use a J48 algorithm to make a classification. J48 is an extension to ID3 and has additional features which compensate for missed values, pruning decision trees, constant value sets for characteristics, rule derivation, etc. This implementation uses the 10- fold cross-validation test mode. The time it took them to build the model is 0.03 seconds, and the accuracy obtained by J48 is 96.3647%.

III. Dataset Description and Preprocessing

The first synthetic dataset, a collection of 13234863 million transactional records spanning the entire year of 2025, was provided by LBG. See figure 3.1:

Dataset statistics	
Number of variables	4
Number of observations	13234863
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	2132448
Duplicate rows (%)	16.1%
Total size in memory	403.9 MiB
Average record size in memory	32.0 B

Variable types	
CAT	2
NUM	2

Figure 3.1: First Dataset Statistics

The second synthetic dataset, a collection of 174601 transactional records spanning the entire year of 2020, was provided by LBG.

Dataset statistics	
Number of variables	7
Number of observations	174601
Missing cells	44093
Missing cells (%)	3.6%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	9.3 MiB
Average record size in memory	56.0 B

Variable types	
NUM	4
CAT	3

Figure 3.2: Second Dataset before preprocessing. There are 4409 null values

Dataset statistics	
Number of variables	7
Number of observations	174601
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	9.3 MiB
Average record size in memory	56.0 B

Variable types	
NUM	4
CAT	3

Figure 3.3: Second Dataset after filling the missing values

We applied the following preprocessing steps:

We made a dataframe for the sender account and the receiver before categorising them by the amount of money and date. In addition, we applied an aggregate function that helped us to produce multiple statistics, based on calculations per amount of money or date in a single calculation. For instance, the 'amount of money' will have ['sum', ‘count'], and 'date' will have [min, max]. Subsequently, we changed the multi-index into a single index to flatten the index. In the modelling section, we used the K-means method to produce a 'cluster' label for all of the customers and applied the labelled data as train set for the decision tree model.

For the second dataset, we added a new column called ‘status’ to check the transaction. This helped us identify whether the transaction had been ‘withdrawal’, meaning money leaving the customer account or ‘deposit’, meaning that money is entering the customer account. We also added a new column called ‘ withdrawal’ to change the amount of money from a negative number to a positive number. This helped us to plot the withdrawal transaction and made it clearer to understand.

IV. Methodology

A. *First Dataset*

The approach followed for our data analysis is as follows:

- First, we applied an exploratory data analysis on the dataset to extract useful insights and to obtain a broad overview of the dataset.

- Second, based on this exploratory analysis, we extended our analysis method to implement an unsupervised learning technique, which is the K-means clustering technique, to group customers, based on their spending density and number of transactions.
- Third, we performed a decision tree classifier.

All the code files are provided in our GitHub repository [5]

1. Dashboard:

Using pandas and gridspec from 'matplotlib' library, we were able to build a dashboard that represents a detailed customer transaction (money entering and leaving the account) within a given time period, as displayed in figure 4.1. The purpose of the dashboard is to investigate customer behaviours through their transactional patterns. The first row in the dashboard has a table and a bar graph representing money transferred from a given customer account to other personal accounts. The second line

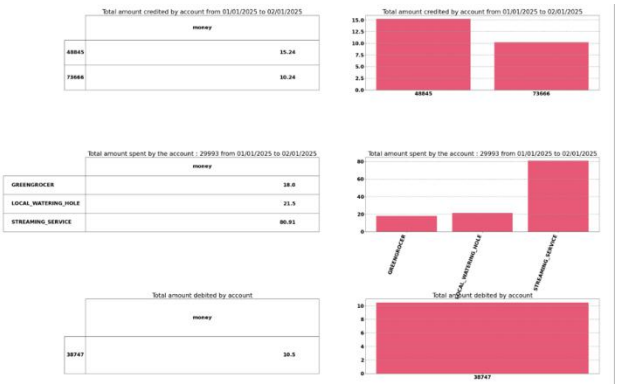

Figure 4.1: Customer Dashboard

represents money spent on shops from a given customer and the third row represents another personal account transferring money to this given customer account.

2. Clustering:

Lloyds Bank's vast databases and voluminous information include customer information and the history of their transactions.

Lloyds Bank could benefit from these databases to increase their profit. This is possible by finding useful ways to analyse their customers' behaviour.

This paper uses the 'K-means' clustering method to analyse and categorise customers in the Lloyds Bank dataset.

We seek to categorise datasets by their characteristics into similar groups of observations, which is also referred to as "clusters". Observations within a given group will be similar to other observations in the same cluster and dissimilar to observations in other clusters, using clustering models.

The basic concept behind K-means clustering method is to find the 'K' average or value at which the data can be clustered, splitting the data set into K "group"s.

We take into consideration the number of transactions and amount of transaction. Evidently, the similarity or closeness of data is grouped into several clusters, each with a high level of similarity among cluster members. Furthermore, we conduct this step to identify the customer segmentation, which helps collect knowledge regarding customer characteristics. One method that could manage the relationship between customers and Lloyds Bank is to offer a different treatment, based on the customer characteristics within each customer segment. Thus, the purpose of clustering is to customise the products, services, and marketing messages for each individual cluster.

3. Semi-supervised learning Modelling:

Model designing:

Top commercial organisations should be able to obtain comprehensive information about their customers, such as product preferences, risk preferences, financial preferences, etc. They also should create quantitative models for business applications, by which they can quickly discover marketing trends and provide customers with differentiated products and service strategies. Therefore, the most important challenge for Lloyd Bank is to make a detailed classification of their customers and provide them with the most suitable financial products or services. In this part of the report, we'll build a classification model by semi-supervised learning and utilise it to automatically divide current customers and new customers into different groups.

Customer classification is useful in identifying the various requirements of different customer groups, as well as the typical characteristics of each customer group and to guide the development of marketing and sales. However, unlike the music or video recommendation engine, the financial products and services do not require personalised recommendations, but only grouped recommendations. Therefore, the recommendation engine is made robust by the classification model and by matching the service to the suitable group.

Under normal circumstances, customers are categorised into groups depending on their current status, taking into account personal aspects such as age, occupation, region, title, education, etc. Furthermore, behavioural attributes are also taken into account, for example, trading behaviour, consumer behaviour, calling behaviour, etc.. In general, banks classify customers according to five principle themes: customer attributes, customer behaviour, customer value, customer needs, and customer attitudes. However, current data does not provide sufficient information about customer backgrounds and their financial preferences, thus, our classification technique was restricted to grouping the transfer record data by the account numbers and making a classification based on two dimensions: the frequency of transaction and the total amount of transaction.

We suppose that the frequency of transaction can reflect the activity of customers, which is equivalent to the customer's loyalty to the bank. The total amount of transaction can show the customer's financial capacity, which can be used to measure the value of the customer to the bank. Therefore, in combining those two features, the customers can be divided into four different kinds accounting to their corresponding characters, which can be seen by following graph Fig 4.2.

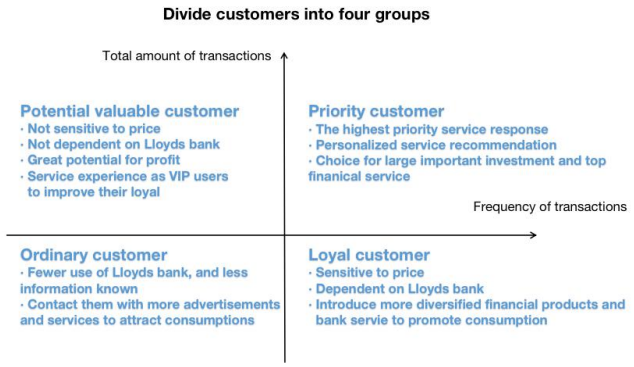

Figure 4.2: Description for each kind of customers

For different groups, we can adopt different sales strategies and recommend different financial services to them.

·Ordinary customer:

'Ordinary customers' do not transfer funds frequently and the total amount of transfers is not considered substantial. They may not have a high demand for use, or they may use multiple banks at the same time. Information for 'ordinary customers' is restricted and they may be potential loyal customers or valuable customers. This type of customer is the key target group for activity promotion or preferential activities and the bank can reach these customers through telephone sales or gifting, to try to improve their loyalty.

The bank may introduce these customers to select, basic financial products, such as like credit loans, mortgages or loan securitisation. However, the loan limit is restricted, because the actual economic capabilities of such users is unknown. Thus the bank may only offer them the opportunity to experience more diversified financial services.

·Loyal customer:

'Local customers' often use Lloyds Bank to make transfers, but the total amount of transfers is considered to be very low, which explains they are loyal to the bank, even though they do not have enough financial strength to make large investments. Banks can use such users as information dissemination sources, prompting them to recommend bank activities and products to people around them, and give them some benefits or discounts as rewards for attracting new customers.

Due to the high loyalty of such users, the bank can open all loan services for them with a medium loan amount. The bank may also recommend a variety of low-risk financial products to them, such as monetary fund, national bonds or a time deposit.

·Potential valuable customer:

'Potential valuable customers' have made high-value transfers through the bank, indicating that they have high asset strength and are not price sensitive. However, the Lloyds bank may not be a dependable choice for them. They have huge potential value, since they are very likely to become priority customers for the bank, and banks need to give them the best experiential service to increase their loyalty to the bank.

Lloyds banks may use professional sales staff to recommend this customer type with the most suitable financial services and investment consulting. The bank may also guide in becoming 'super customers' for the bank, which may encourage them to invest more money into the bank. However, for loan services, the banks can only give them a medium loan amount, since their credit for the bank may still be uncertain.

·Priority customer:

Such customers should be the top priority for the banks and they typically possess a high financial capacity, as well as loyalty. These customers determine the development status of the bank. Their needs should be responded to first and they should be prioritised for key investment projects by bank.

The bank should establish personalised investment strategies for these customers, providing them with the most suitable asset allocation schemes and financial planning. Therefore, the bank should understand the needs of these customers, their consumption preferences, credit limit, risk tolerance, asset allocation preferences, risk preferences and other information in order to achieve accurate product recommendations.

B. Second Dataset

For the second dataset, we used 'Tableau' software, which helped us render our visualisation for the data analysis more dynamic and interactive. Hence, the benefit from that we can take an insight across bank-level 'all costumers' or using the filter to select particular customer or shops. The dashboard will present five different graphs. The first graph depicts 'monthly deposits to customers accounts', the second presents the 'monthly transaction per customer', the third graph shows the 'places customers spend their money the most', the fourth graph displays 'hourly deposit transactions to customer accounts' and the fifth graph shows 'hourly withdrawal transactions per customer'. See figure 4.3 .

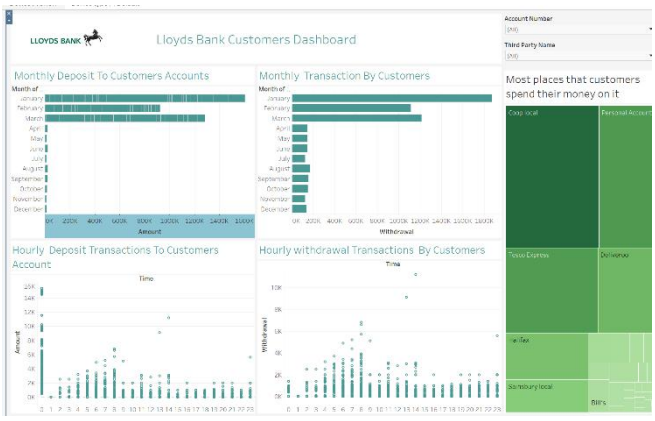

Figure 4.3: Lloyds bank customers dashboard

We have attempted to apply the same semi-supervised learning method to train the classification model. However, the second data set is not large enough its distribution is uneven. Therefore, we decide to use the first data set to train the model, before using the model to make predictions based on the customers of the second dataset.

V. RESULTS

We started an exploratory data analysis for the bank level before expanding into the details of the customer level.

A. *Exploring the dataset:*

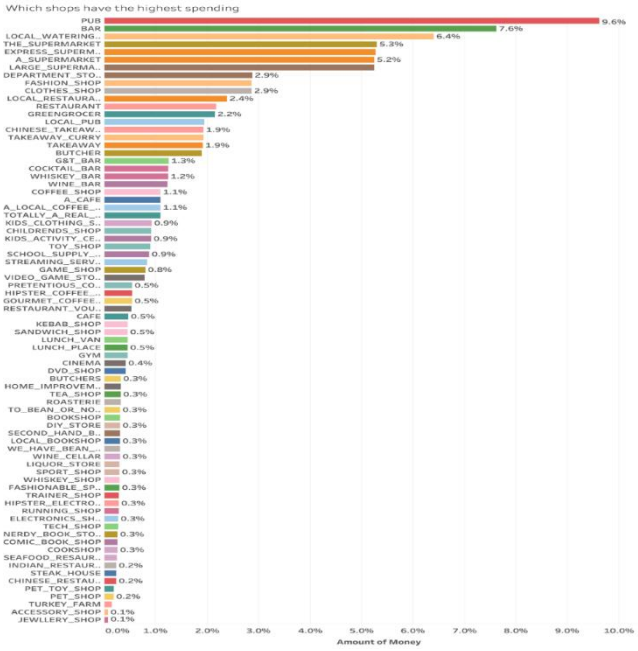

Figure5.1: Which shops have the highest spending density?

From figure 5.1 we can see that 9.6% (the total amount of money that the pubs received for the year 2025 is £5,348,388) of the customers spending in this sample dataset goes to pubs and 7.6% (with a total of £4,237,389) goes to bars. It is possible for LBG to investigate and evaluate this with their real customer data.

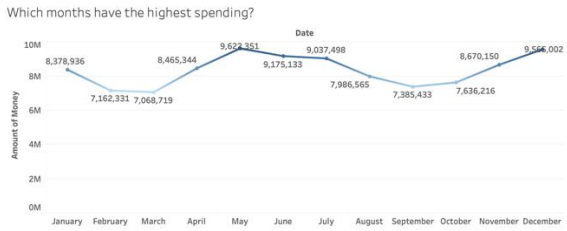

Figure 5.2: Which months have the highest spending density?

Figure 5.2 indicates that spending increases at the start of the year (January) and at the end of the year (December). Also, we notice that spending increases at the end of spring (May). On the other hand, spending decreases at the end of summer (August).

B. *Data clustering and labelling with K=4:*

After defining the categories, we started the semi-supervised learning process of classification. Since the original data was unlabelled, we needed to use the 'K-means' method to apply clustering on the data and to form category labels. In turn, we used the data as a training set to train the decision tree algorithm. After the training of the model, we can use it to classify new customers, based on its records, in a month or half of a year.

The first step is to group all of the records by the senders, and count 'the frequency of transaction' and 'the total amount of transaction' for every customer as two features for the train set. Before the clustering step, we'll make a scatter plot to show the distribution of all the customers:

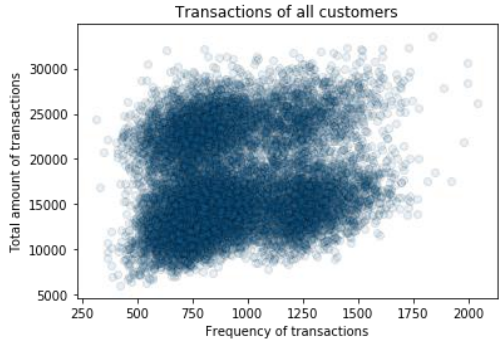

Figure 5.3: The distribution of all the customers

From figure 5.3, we can identify that all of the points are gathered into four piles, which indicates that all of the customers can be clustered into four groups. We can test this hypothesis by applying the elbow principle, which determines how many centres should be set for the K-means method.

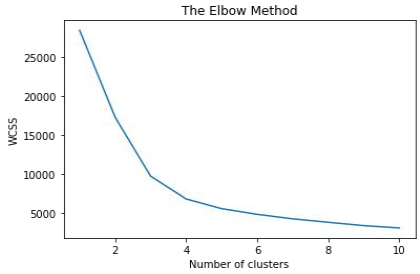

Figure 5.4: Elbow principle for choosing the number of clusters

As we can see from figure 5.4, there is a turning point at number 4 and 5, proving the conformity of the data with our model. Therefore, we can cluster the data into four groups and put the 4-categories labels on the data. The result of the clustering and the labels of data are showed as below:

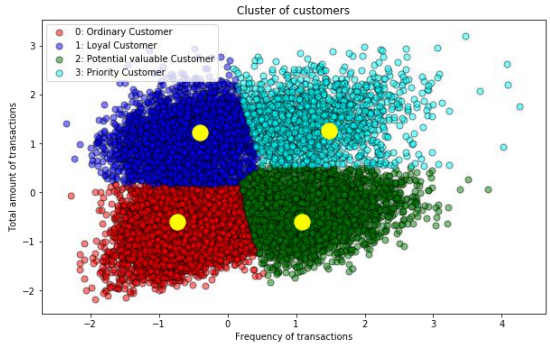

Figure 5.5: The four cluster of all customers

In this clustering, we labelled the four kinds of customers as: 0 - ordinary customer, 1 - loyal customer, 2 - potential valuable customer, 3 - priority customer. We used these labels and data as a training set to train the classification model in ext part. The number of customers in each cluster are: 6318, 3326, 3089, and 1517, respectively. This indicated that approximately half of the customers had been ordinary customers and that Lloyds Bank should invest more in its marketing strategy for ordinary customers.

We also gained some insight into the four groups of customers. Customers who were classified as potentially valuable customers and priority customers tended to spend 70% more than ordinary customers and loyal customers. Therefore, the bank can achieve almost twice the profit if it maintains the stability of these customers. Meanwhile, loyal customers and priority customers' frequency of transaction

was 1.6 times that of ordinary customers. This suggests that they are more likely to accept more types of financial products and services. Some statistical information is displayed in the graph below:

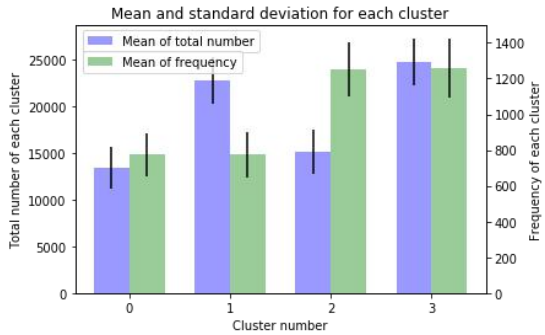

Figure 5.6: Mean and standard deviation values for each cluster

C. *Decision tree classification model*

Upon identifying that the customers' different financial profiles can assist the bank in evaluating customer expected spending and prospective income, we built a classification model. After the clustering, we split them into a training set and a testing set. Then, we used those two sets to train a decision tree model. The decision tree classifier is based on 'Gini impurity', which can be used as an alternative to entropy. If there are k classes and, in a given branch, B, and the proportion of training data of class I is Pi, then the Gini impurity for that branch is given by the following formulation:

$$G(B) = 1 - \sum_{i=1}^{k} p_i^2$$

Before we build the tree, it is necessary to create methods to deal with the over-fitting and under-fitting problems. There are various constraints that we can put on the decision tree which can affect over-fitting and under-fitting of the data, and the simplest is restricting the depth of the tree. We use the cross-validation method to choose the best number of the depth of the tree. As the chart below depicts, the optimum depth of the tree is eight.

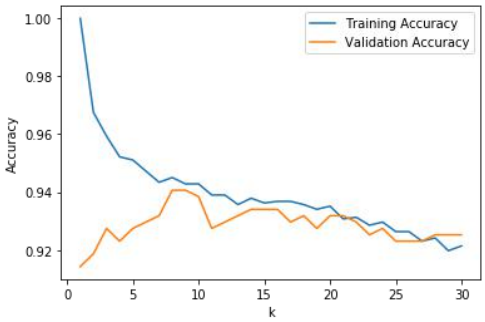

Figure 5.7: Cross-validation for choosing the depth of decision tree

One of the advantages of decision tree classifiers is that they are very interpretable. This is because the tree corresponds to a set of 'if-then' rules, which identify the class of an input, given various constraints on the features. The structure of these rules can be seen by visualising the tree. The structure of the decision tree for our data is shown as below:

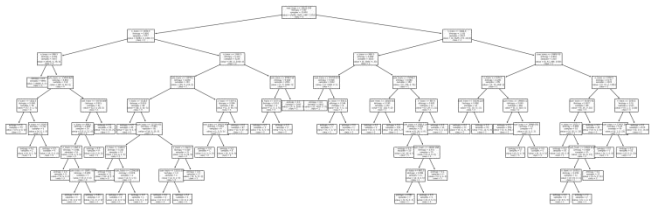

Figure 5.8: The structure of the decision tree for the first data set

20% of the test set suggests that the accuracy of the decision tree model is:

training accuracy=0.99973, test accuracy=0.99719.

Thus, we decided to utilise the model from the first data set to make a prediction on the customers of the second data set. The result of prediction is displayed below. We can see most of the customers in dataset two are classified as ordinary customers; potential valuable customers are very rare. Compared to data set one, this data set is incomplete, or some potential users have not been identified. Therefore, the bank should focus more on ordinary customers and encourage them to become more active, by implementing corresponding affective marketing strategies.

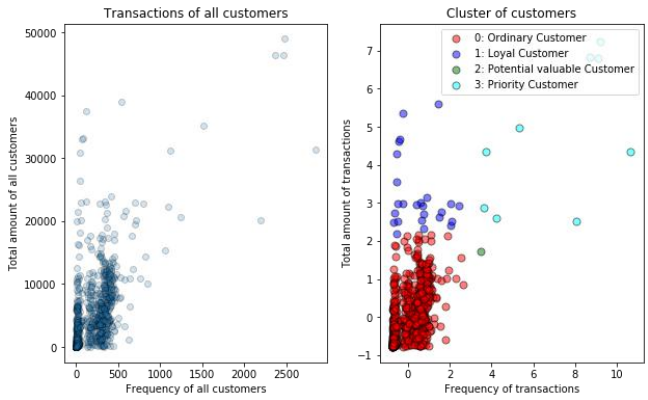

Figure 5.9: The classification result on the second data set

With the decision tree model, Lloyds Bank is able to make classifications on current customers and future , prospective customers, as long as they are able to obtain the customer's behaviour data for several months. Equipped with the 4-categories model, the bank can also make detailed classifications of their customers and provide them with the most suitable financial products or services.

D. *First dataset dashboard*

Statistics suggest that online and bank fraud is increasing. To reduce fraudulent transactions, Lloyds Bank could develop a fraud risk measurement software for clients.

Since the data isn't labeled, fraud possibility is detected by identifying anomalies. In this case, the outliers of the data in the clusters are suspected to have a fraud possibility and need to be audited. Here, we calculated the distances from centroids. Then, the values that exist after 95 percentiles of the data count as outliers. We checked some individual account behaviour from these outliers in the built-in dashboard that we made earlier.

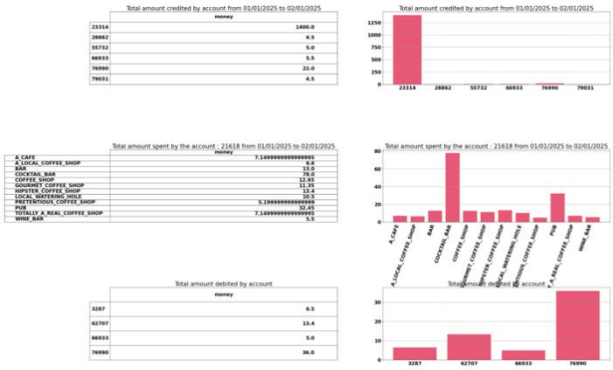

Figure 5.11 (a): Investigating Outlier Customer Num

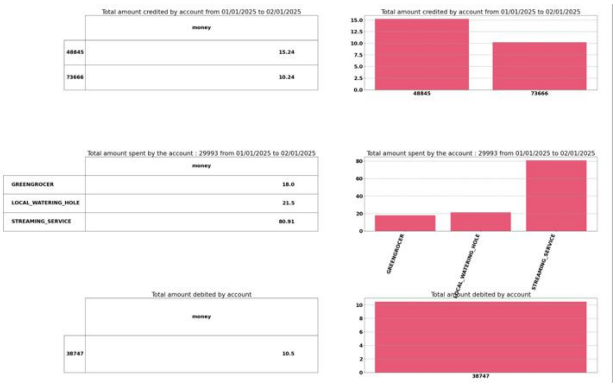

Figure 5.11 (b): Investigating Outlier Customer Num

Outliers' customers were included in our clustering method. Hence, we investigated their summary transactions through the dashboard built previously. Figure 5.11(a) and 5.11(b) suggests that, from 01/1/2025 to 02/01/2025, the account 29993 spent £80.91 on the streaming service alone. We identified this as potentially suspicious behaviour, compared to their daily spending for each transaction had ranged between £10.24 to £21.5, a similar scenario for the account 21618. These accounts contained unusual transactional behaviour and it would be worth calling them by telephone or making contact by any other means to ensure that the transactions are valid.

E. Second dataset

At this stage, we used 'Tableau' software to build the Lloyds Bank customers' dashboard to provide an insight into the graphs on the dashboard. In addition, this dashboard facilitated the understanding of customer behaviour and will present suggestions regarding our observations and illustrate what benefit the bank might gain from the customer behaviour.

The dashboard shown in figure 5.12, without any filter [6].

As shown in figure 4.3, the '*monthly transactions by customer'* graph suggests that, overall, monthly customer spending reached a peak in January (1,870,245). This is assessed as likely to be caused by the new year holiday season, during which consumer shopping and travel increases. Thus, bank advertisers take this opportunity to deliver bank products, such as different type of credit cards, and currency cards, at this time of the year.

The '*monthly deposit to customer accounts'* graph shows that the money that entered customer accounts hit a peak in January (1,604,459). It is important to reiterate here that consumers activity increases in January and there is a likelihood that business customers may observe an increase in their profits during this month. Furthermore, customers may also receive their annual bonus in January. The bank can subsequently align messages to consumer behaviour and strategically send them messages about investment offers during the same month.

The graph also shows hourly *'deposit transactions to customer accounts'*. The graph suggests that the money going into the customers' accounts at 12:00 am were larger deposits compared to that of other hours. This larger than average sum may be the customer's salary. This graph may help banks to investigate if there are any outliers and to investigate or flag up the event if it is assessed to be a fraudulent transaction.

On the other hand, the '*hourly withdrawal transactions by customers'* graph shows that customers tend to spend more money from 08:00 - 10:00 am. This figure helps to identify any outliers and prompts the necessity to check any suspicious transactions.

As mentioned in the first dataset analysis, the bank reap huge benefits by understanding their customers' behaviours. For instance, where do customers most spend their money? Here, we are able to obtain specific details about the different places customers spend their money. The graph suggests that the place that customers spent their money the most was in Coop local (1,591,417). This information is beneficial to the bank and allowed it to identify the need to create a new deal with Coop local. As a result, customer satisfaction increased due to the new offers in their favourite place and that also increased their loyalty to the bank.

Now, let us take a look at specific customer behaviour at Lloyds Bank. We used the filter and selected a customer with the account number: 1007.

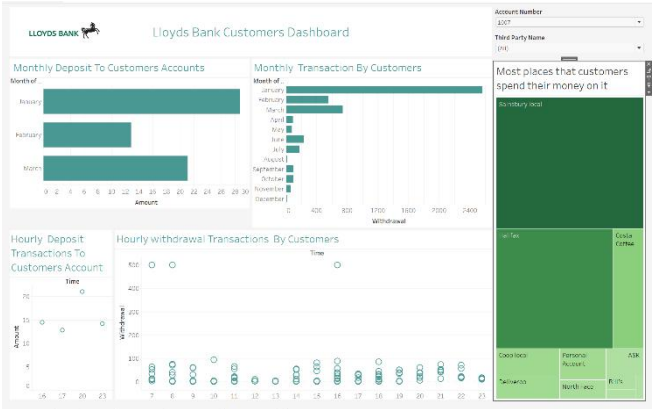

Figure 5.12: dashboard for account number:1007

Figure 5.12 depicts the graph showing *'monthly deposit to customers accounts'*. The graph indicates that this customer received several deposits throughout the year. A detailed analysis of the account suggests that in January, the customer received 28.70 GBP, in February 12.83 GBP and in March, 21.07 GBP. The customer received this money from another personal account. The analysis indicated that this customer did not receive a monthly salary.

The graph '*monthly transaction by customer'* indicates that the customer spent a small amount of the money over the course of the year. The highest amount of money they spent was in January, amounting to 2,564 GBP.

The graph titled *'hourly deposit transactions to customer's account'*, indicated that this customer received their money from 16:00 - 23:00 PM, on four separate

occasions during the course of the year. The money had originated from a personal account. In contrast, this customer spent the money from 07:00 am - 23:00 pm. The graph titled '*hourly withdrawal transactions by customers'* indicates that the customer spent small sums of money in each transaction. Furthermore, places that this customer spent their money the most were in supermarkets or coffee shops. This indicates that this customer could be a student.

Furthermore, this dashboard provides beneficial data and insights by representing transactions and core customer information. For instance, it allows us to identify whether the customer has a salary, which may help Lloyds Bank in identifying the life stages and current statues that allow for better product services and targeted marketing for their customers. The dashboard would assist Lloyds Bank to target customers with the most appropriate products, based on the data, at varying stages of the customer's life.

VI. Conclusion and Future Work

Banks generate massive amounts of transactional data as a consequence of the day-to-day operation. We presented a method to facilitate the analysis process of large-scale transactional datasets, using both supervised and unsupervised learning techniques. Our case study contains two large synthetic datasets. The first dataset has 13234863 million transactional records for the year 2025 and the second dataset has 174601 transactional records for the year 2020, which we analysed using 'K-Means' clustering and decision tree classification algorithms. The motivation for this work is to make use of Lloyds Bank sample datasets effectively and promote a broad variety of possible approaches and potential solutions to meet Lloyds Bank's customers' needs.

With the rapid increase of data and the advancement of modern technologies, banks management must adopt and adapt to new strategies that keep up with the pace of this evolution. In this research, which is an analysis of sample datasets provided by LBG, we have been able to understand customer behaviour and classify it according to their customer's common purchasing characteristics. We also shed light on several account numbers we assessed to contain suspicious patterns in their money spending behaviour, based on their transactional data. The analysis approach that we took from clustering customers and finding outliers customers could be a helpful analysis approach to follow for Lloyd's Bank to detect fraud with their real data.

The next step in our research aims at investigating other ways of expanding our analysis, such as building a regression model that predicts the balance of a customer at the end of the month. This will be based on a customer's daily spending and balance. We also plan to investigate the issue of customer churn with the datasets that we have. For example, checking classified customers in our analysis and identify inactive customers that could be a sign of a churn. In addition, since the data is quite structured, supervised machine learning would be applicable. Linear regression, logistic regression and decision trees can be applied. first, classifiers would have to be developed. These classifiers can be the spending density patterns of the account numbers. Neural networks can also be implemented to build prediction models. The data would have to be split into training and testing sets. The model would have to build on the training set. Moreover, we plan to deal with the issue of the dimensionality of data. When implementing a neural network, the low dimensional features of the data set can be extracted by a 2D convolution of the spending density patterns and the places that the spending has occurred. On the other hand, a 2D cross-correlation of the spending density patterns and the places that the the spending has occurred would give a relationship between the classifiers, i.e., the spending density patterns.

VII. References

1. Wei, W., Li, J., Cao, L., Ou, Y. and Chen, J., 2013. Effective detection of sophisticated online banking fraud on extremely imbalanced data. *World Wide Web*, *16*(4), pp.449-475.
2. Bholat, D., 2015. Big data and central banks. *Big Data & Society*, *2*(1), p.2053951715579469.
3. Chitra, K. and Subashini, B., 2011. Customer retention in banking sector using predictive data mining technique. In *ICIT 2011 The 5th International Conference on Information Technology*.
4. Subia, P.M.G. and Galapon, A.C., Sample Model for the Prediction of Default Risk of Loan Applications Using Data Mining.
5. https://github.com/AliPythonista/DSMP-Group-27
6. https://public.tableau.com/profile/ghadah.almuqrin#!/vizhome/LloydsBankCustomerDashboard/Dashboard1